

Portfolio/presentation: <https://github.com/Alessandro266>

Project Title

Incident Response Exercise: Detecting and Analysing a Malware file on TyHackMe

Brief Summary

In this exercise I completed a hands-on incident response task on TyHackMe to learn the steps involved in detecting and handling potential malicious files. The challenge focused on investigating a suspicious file and analysing its behaviour on a windows system in which I followed a series of steps to delete the file safely.

Through this process, I learned how cybersecurity professionals identify malicious activity, verify threats and restore system integrity. This experience deepened my understanding of system monitoring, malware and the overall incident response process, while also improving my ability to connect theory with real-world cybersecurity scenarios and practices like this one.

Introduction

TyHackMe is an online cybersecurity training platform that offers interactive guided labs to help people learn practical skills in areas such as incident response, malware analysis and threat detection. The platform is designed to provide a safe environment for practicing real world cybersecurity tasks and making it an ideal choice for students like me.

The reason I chose TyHackMe for this project was because it allows hands on experience with realistic scenarios and virtual machines. The incident response room, provided me an opportunity to explore system monitoring, identify suspicious files and understand the steps involved to respond to potential threats.

Problem/Challenge

The scenario in this exercise involved investigating a user's workstation that experienced severe performance issues and high CPU usage. Initial checks revealed continuous connections to a single IP address despite no alerts from SIEM. This indicates a potential compromise or hidden malicious process. The main challenge that was at hand in this exercise was to investigate a subspinous file and determine whether it was malicious by following standard incident response procedures.

Project Goal/Objectives

The goal of this project was to investigate a potential security incident involving a compromised workstation and demonstrate the incident response process.

Furthermore, the project's objectives are the following:

- Examine system processes and files to detect potential malware
- Apply appropriate incident response steps to contain and remove the threat
- Document my findings and reflect on how incident response tools and techniques support real world cybersecurity operations
- Identify abnormal system behaviours and determine its main cause

Methodology

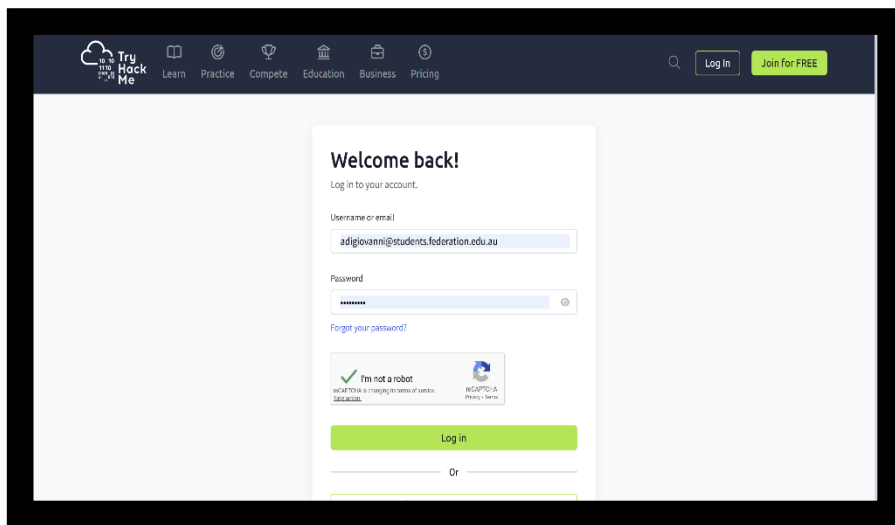
As I read through the scenario, I launched the virtual machine which simulated a real-world security investigation. The first step involved me observing a system performance through task managers where I identified an unfamiliar process consuming high CPU resources. I then found the IP address using the command prompt's directory code which TryHackMe gave me. Moreover, I checked the users Microsoft edge downloads and discovered a suspicious file with the extension .docm (a macro enabled word doc)

After opening this file, I noticed a link embedded within it and confirmed the presence of a macro by viewing it under the "view macros" tab in the word document. This confirmed the files malicious intentions.

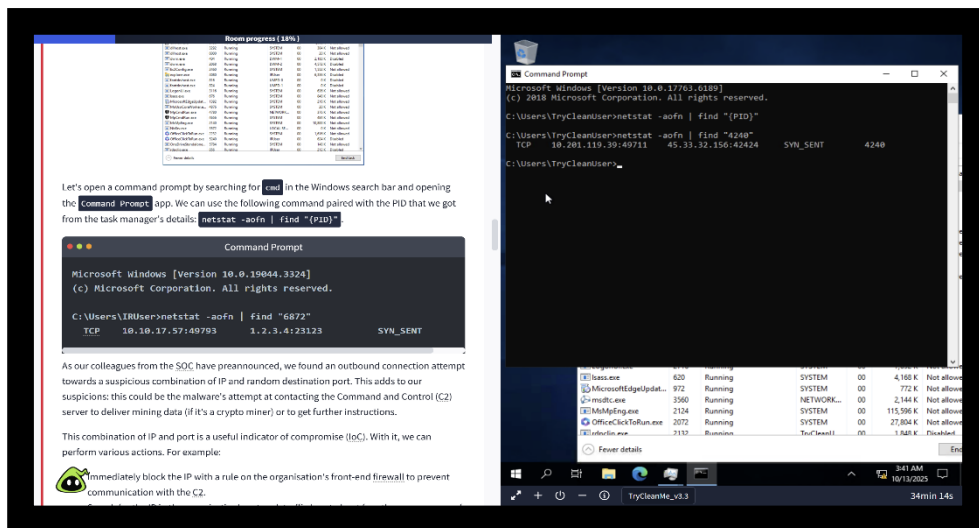
Next, I examined the windows run registry key that TryHackMe provided me to identify suspicious activity. Here I found a suspicious entry name "default app" which highlights that if the user closed their computer the malicious file would be set to open immediately after the computer is logged into on startup.

Ultimately, I right clicked over the entry and deleted it and the related process in task manager, completing the incident response procedure effectively and successfully.

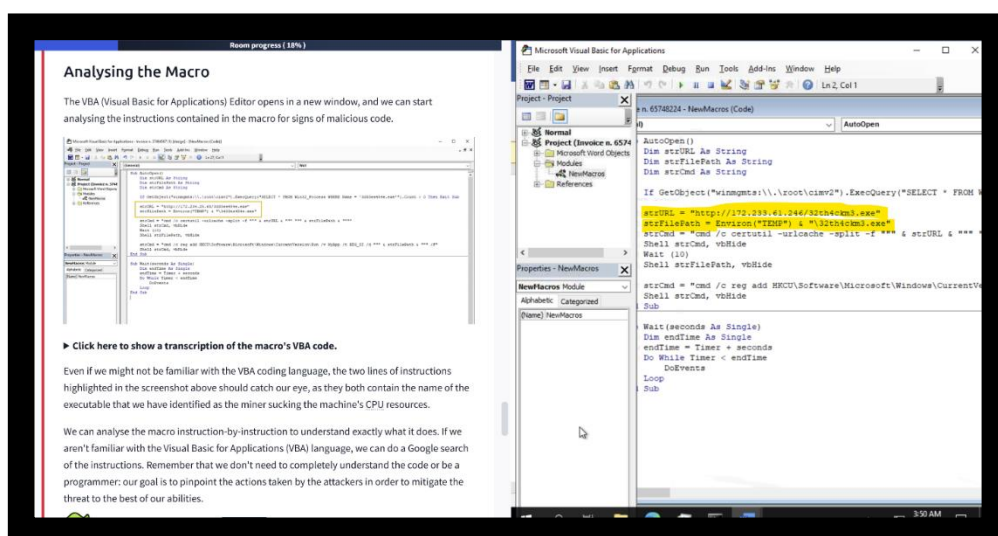
Results/Outcomes



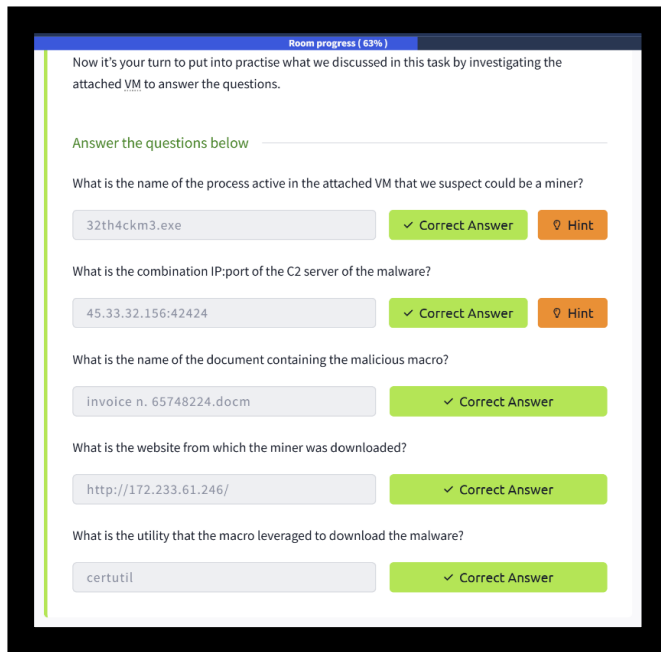
Screenshot 1: Proof of identity
(student email)



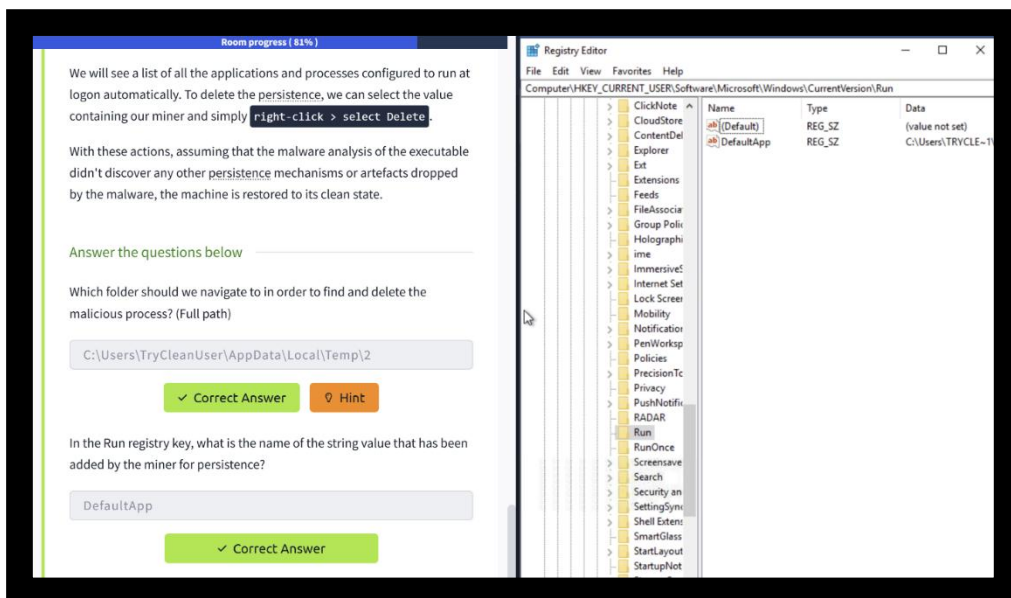
Screenshot 3: Finding the external IP address using directory codes



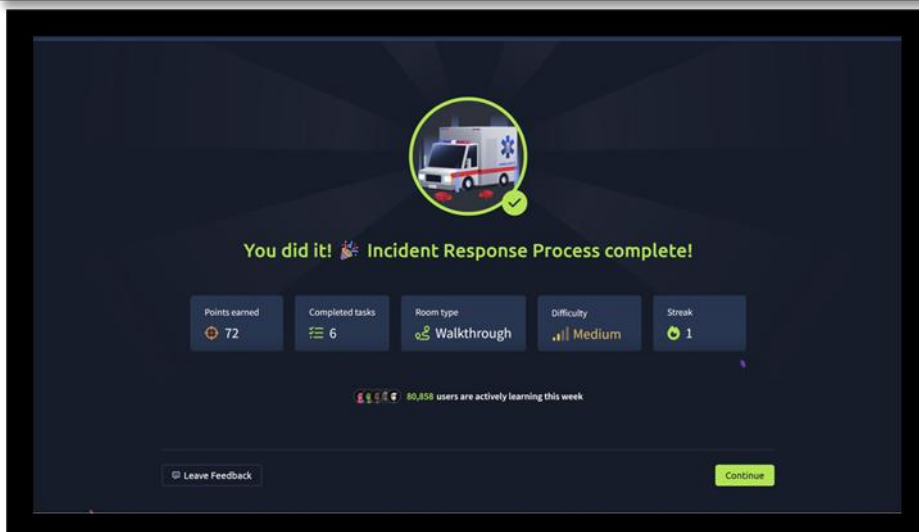
Screenshot 4: Finding the macros VIA “view macros”



Screenshot 5: Answering the questions



Screenshot 6: Identifying what has been added to the registry



Screenshot 7: Proof of completion

Reflection

What did you learn about cybersecurity concepts and tools from this exercise?

I learned how to identify and analyse malicious, detect suspicious registry entries and basic forensic tools such as task manager and Microsoft word when looking for macros. This exercise strengthened my understanding of incident response, particularly the detection and containment stages

- **How do you see this experience contributing to your professional growth as a future cybersecurity practitioner?**

This activity helped me apply theory to practice and overall boosted my confidence in analysing unusual system behaviours. The skills I gained from this exercise are relevant to working in a security operations centre (SOC) and an incident response team, where quick investigation and threat protection practices are essential.

- **Looking back, what would you do differently if you repeated the task?**

I would document my steps in more detail as I often found myself flowing through the instructions without taking more screenshots of critical information. Additionally, I would also spend time examining related system and network logs to understand how the malicious file established connections and affected system performance. This would help me build a more complete picture of the incident and build my ability to handle similar cases in a professional setting.